Rental-Implied Depreciation of AI Accelerators: Why the Estimates Disagree

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Abstract

How fast do AI accelerators lose value? Published estimates range from roughly flat to about 40% a year. This paper rebuilds the question from posted rental prices: a panel of 24,210 listing-months of accelerator rents from October 2017 to November 2026, centered on seven NVIDIA chips from the V100 to the B200, across hyperscalers and neoclouds and across on-demand, spot and reserved contracts. Because age equals calendar time minus vintage, the age profile is not identified with chip and month effects, so the paper reports each chip's realized price path and four identification strategies side by side. Three findings stand out. First, realized rents fall slowly: the V100's rent changes by -0.38 log points over nine years. Second, the shape of the age profile depends on the market read: spot prices fall late, while list prices that actually change fall early, and 96% of month-to-month on-demand and reserved transitions show no change at all. Third, with valid few-cluster inference no strategy identifies a significant decline or shape. Turning rents into asset values, rents stay far above power and maintenance costs for the whole observed life, so economic life is set by obsolescence and power scarcity rather than by rents reaching operating cost. The disagreement in the literature is largely a disagreement about which price series and which identification assumption to use.

1 Introduction

Reported earnings at the largest AI spenders now hinge on one management estimate: how many years a server lasts. Between 2020 and 2025 Microsoft, Alphabet, Amazon, Meta, Oracle and CoreWeave lengthened the useful lives of their servers, and Amazon partly reversed course for a subset of them in 2025 (Section 2 and Table 1). Whether five- or six-year straight-line lives are defensible for graphics processing units (GPUs) bought at the peak of an investment boom is now a live question for investors, auditors and regulators.

The first academic and practitioner estimates of accelerator depreciation have appeared, and they disagree. Amissah-Ocran and Ayumu (2026) fit a rent–age profile at one date and find a retention

*Virginia Commonwealth University. Working paper; comments welcome. Replication package: see the data and code availability statement. Use of AI tools: the code, data collection, analysis, figures and drafts of this paper were produced with Claude (Anthropic), running in Claude Code, under the author's direction; see Appendix C.

path consistent with roughly 40% a year; Chincholikar and Chawla (2026) use executed resale prices and find 24–27%; Huang (2026) measure rental contribution on a peer-to-peer marketplace and find 17–32%; and archived posted on-demand prices show declines from roughly 15% a year to none at all (Compute Credit Index Research, 2026). These studies measure different things: resale value, net cash flow or posted rent, at one date or along a path.

This paper puts those measurements on one basis, using the object that is easiest to observe over time: what it costs to rent an accelerator for an hour. I build a panel of 24,210 listing-months of posted rents covering October 2017 to November 2026; the estimation sample is 20,185 listing-months for seven NVIDIA chips, from the V100 (2017) to the B200 (2024). The panel combines Amazon Web Services (AWS) price-list history, archived pricing pages, the SkyPilot catalog history and live collection, across hyperscalers and neoclouds and across on-demand, spot and reserved contracts (Section 3).

The central econometric problem is old. Age equals calendar time minus vintage, so with chip and month fixed effects the linear part of the age profile is not identified (Hall, 1968). I therefore report each chip's realized price path first, which needs no identification assumption, and then run four identification strategies side by side: a quality-adjusted vintage effect, a frontier-relative obsolescence term after Geske et al. (2007), a successor event study, and a shape-only specification that sweeps an assumed linear rate (Section 4). Inference is at the chip level, which leaves seven clusters.

Three results stand out.

Realized rents fall slowly. Holding provider, region and contract fixed, the V100's rent changes by -0.13 log points over its first four years and -0.38 over nine; the A100 40GB's falls to -0.45 by year five. The H100's rent rose by 0.24 in its first year and was -0.50 at three years, before 2026's shortage brought it back to -0.04 at four years.

The shape of the age profile depends on the market, and the data mostly measure pricing habits. 96% of month-to-month on-demand and reserved transitions in the panel show no price change; AWS has charged $3.06 an hour for a V100 on demand in every year from 2017 to 2026. Spot prices give a back-loaded profile (rate over years 0–2 minus the rate over years 2–4 of 0.87), while list prices that do change give a front-loaded one (-0.52, provider-clustered standard error 0.24). Which of these is "the" age profile is a choice of market, not a technical detail.

With valid inference, the data do not pin down the rate or the shape. Under the unrestricted wild cluster bootstrap that the analysis specified in advance, the pooled shape estimate (0.47) excludes zero. A Monte Carlo on this panel's design shows that the unrestricted bootstrap's 95% intervals cover the truth only 66%–72% of the time, while the restricted version covers about 98%; under the restricted bootstrap the shape has $p = 0.29$ and a 95% interval of $[-0.44, 1.03]$. No identification strategy that identifies a linear rate finds a significant decline over years 0–4.

Turning rents into asset values (Section 5.5), the present value of net rent falls along the realized path by 32% by year three for an H100, which is slower than any five- or six-year straight-line schedule. The H100's neocloud rent of $4.50 an hour compares with roughly $0.30 of power and

maintenance, so rents never fall to operating cost within the nine-year horizon: economic life is set by obsolescence and power scarcity, not by rents reaching operating cost. A separate benchmark, term-structure and retirement analysis (Section 5.6) places the power-constrained retirement age at 4.1 years and the life to operating cost at 9.8–11.7 years, with book lives in between.

The contribution is a reconciliation. The spread of published rates comes mostly from three choices that are rarely stated: which price series is read (resale, spot, posted list, or transacted rent), which identification assumption pins the linear age term, and whether value is taken to end when rent reaches operating cost or when a better chip displaces the old one in a power-constrained slot. The paper's open panel, code and identification range let readers make those choices explicitly.

Related literature. The paper applies the vintage-price approach to depreciation (Hulten and Wykoff, 1981; Oliner, 1993; Doms et al., 2004; Geske et al., 2007) and hedonic price measurement (Rosen, 1974; Berndt and Griliches, 1993; Triplett, 2006) to a new asset class with an active rental market, as Gavazza (2011) does for aircraft. Its identification problem is the age–period–cohort problem of Hall (1968) and Deaton (1997). On inference with few clusters it follows Cameron et al. (2008), MacKinnon and Webb (2017), Webb (2023) and MacKinnon et al. (2023). On the accounting side it relates to Keating and Zimmerman (1999). Recent work measures cloud prices (Byrne et al., 2018; Sawyer and O'Bryan, 2023) and GPU price-performance (Hobbhahn and Besiroglu, 2022), and a fast-growing practitioner and working-paper literature argues about accelerator lives (Dutta and Patatoukas, 2026; Rosenzweig, 2026).

2 Institutional background

2.1 Useful lives in financial statements

Under U.S. GAAP a firm depreciates property and equipment over its estimated useful life, and a change in that estimate is applied prospectively as a change in accounting estimate. Servers and network equipment are reported inside property and equipment, and no firm in the sample reports GPUs as a separate asset class. Table 1 lists the server useful-life policies and changes disclosed by the seven firms studied here, each with the filing it comes from (`docs/useful_lives.md` holds the exact quotations and accession numbers).

The economic stakes are large because server capital expenditure has grown fast: a one-year change in the assumed life of a cohort of servers moves annual depreciation by billions of dollars at each of the largest spenders (Section 6).

2.2 How accelerators are rented

Accelerators are rented by the hour in three main ways. Hyperscalers (AWS, Microsoft Azure and Google Cloud) post on-demand list prices for every region and instance type, discounted prices for one- and three-year commitments, and interruptible spot prices. Neoclouds (here Lambda, CoreWeave and Paperspace) post simpler price lists, often per GPU-hour. Marketplaces match

Table 1: Server useful-life policies and changes disclosed in SEC filings

Firm	Fiscal year of effect	Server useful life (years)
Microsoft	2021	$3 \rightarrow 4$
Microsoft	2023	$4 \rightarrow 6$
Alphabet	2021	$3 \rightarrow 4$
Alphabet	2023	$4 \rightarrow 6$
Amazon	2020	$3 \rightarrow 4$
Amazon	2022	$4 \rightarrow 5$
Amazon	2024	$5 \rightarrow 6$
Amazon	2025	$6 \rightarrow 5$
Meta	2022	$4 \rightarrow 4.5$
Meta	2023	$4.5 \rightarrow 5$
Meta	2025	$5 \rightarrow 5.5$
Oracle	2023	$4 \rightarrow 5$
Oracle	2025	$5 \rightarrow 6$
CoreWeave	2023	$5 \rightarrow 6$
Nebius	2026	$4 \rightarrow 5$

Source: 10-K and 20-F filings on SEC EDGAR; quotations and accession numbers in the replication package.

owners and renters at transacted prices but restrict automated collection, so they are not collected here. Large customers negotiate private prices that are not posted anywhere; this paper is therefore about the posted-price market, and Section 7 discusses what that implies.

Posted prices differ in how often they move. Hyperscaler on-demand and reserved list prices are revised rarely and in large steps, while spot prices respond to spare capacity. The age profile one reads depends on which of these series is used, which is the theme of Section 5.2.

3 Data

3.1 Sources

The panel combines four kinds of source, each archived as raw files with the source URL, retrieval time and HTTP status. (1) AWS's public price-list offer files, which give on-demand and reserved list prices roughly monthly since 2015. (2) The SkyPilot project's public catalog history, which records AWS, Azure, Google Cloud and Lambda list and spot prices since mid-2022. SkyPilot's catalog carries no licence, so it is used but not redistributed. (3) Archived pricing pages for Lambda, CoreWeave and Paperspace from the Internet Archive's Wayback Machine. (4) Live daily collection from October 2026 (the Azure Retail Prices API, AWS offer files, Lambda's pricing page and SkyPilot). Chip specifications and release dates come from Epoch AI's hardware database (Epoch AI, 2026), electricity prices from the U.S. Energy Information Administration, and the deflator is the BLS CPI-U (U.S. Bureau of Labor Statistics, 2026). Vast.ai, RunPod and Oracle restrict automated collection, so nothing is collected from them.

3.2 Construction

The unit of observation is one posted price for a provider, SKU, chip, GPU count, interconnect, region, contract type and term in one month, in U.S. dollars per GPU-hour; multi-GPU instance prices are divided by the GPU count. From 61,873 raw rows, zone-level spot prices become one regional median per day, daily observations become one monthly median, and where two collectors record the same offer in the same month the more direct source is kept (the two agree to a median gap of zero). The result is 24,210 listing-months from October 2017 to November 2026. Real prices are in 2025 dollars. The estimation sample drops the L40S (no release date), the MI300X (one month), outlier-flagged prices and the partial month of November 2026, leaving 20,185 listing-months for seven NVIDIA chips (Table 2).

Table 2: Estimation sample by chip

Chip	Listing-months	Providers	First month	Ages (months)	Hyperscaler/neocloud	Max/min provider
V100	5,260	6	2017-10	3–111	1.61	3.10
A100 40GB	5,510	6	2020-11	6–77	2.34	2.67
A100 80GB	3,081	6	2022-05	18–71	1.92	2.47
H100 PCIe	355	3	2023-01	4–49	2.66	1.95
H100 SXM	4,831	6	2023-01	4–49	3.10	3.52
H200	664	4	2024-12	1–23	0.73	1.73
B200	484	4	2025-06	7–23	1.54	2.08

Listing-months in the estimation sample; ages in months since the Epoch release month. "Hyperscaler/neocloud" is the median ratio of median on-demand prices in months where both provider types list the chip; "Max/min provider" is the median ratio of the most to the least expensive provider's median on-demand price.

3.3 What the raw panel shows

The panel is mostly hyperscaler list prices: hyperscalers account for 94% of listing-months and Google Cloud alone for 62% of the estimation sample, because it publishes every region and machine shape. Neocloud history (1,538 listing-months) comes from archived pages. Three features of the raw data matter for everything that follows.

Prices barely move. 96% of the 9,282 month-to-month transitions in on-demand and reserved listings show no price change; AWS spot prices, by contrast, are unchanged in only 17% of transitions.

Levels differ widely across providers. For the H100 SXM the median hyperscaler on-demand price is 3.10 times the neocloud price, and the most expensive provider's median is 3.52 times the cheapest's in a typical month (Table 2, Figure 1).

Raw prices rise with age for the newest chips (Figure 1, right), because age moves one-for-one with calendar time, with the 2023–26 shortages and with which providers list a chip. None of this is an age effect, which is why the paper does not read age profiles off raw prices.

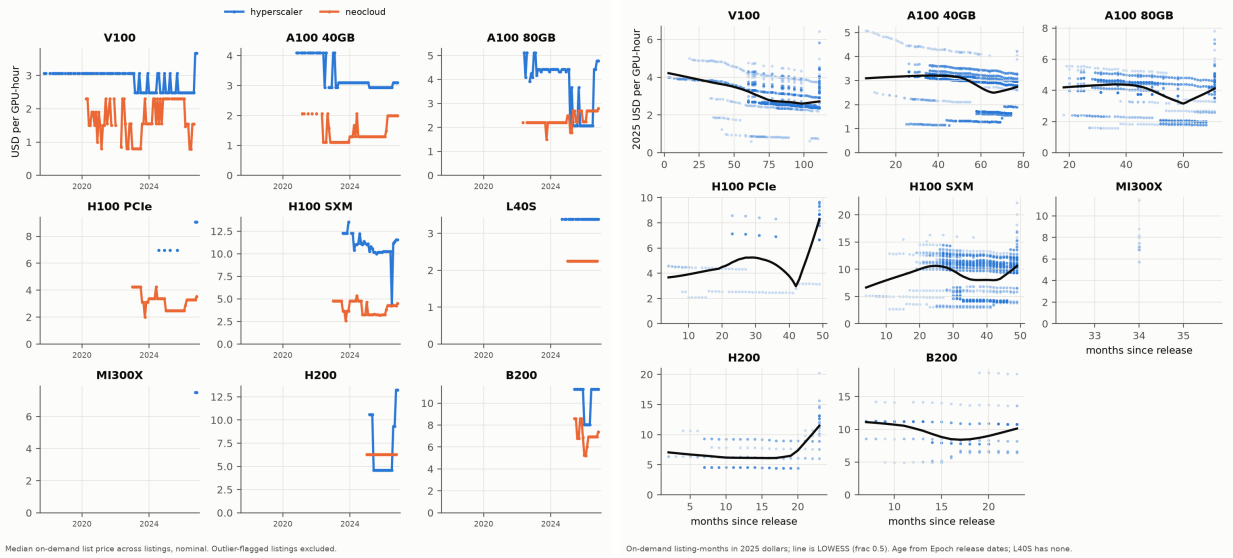


Figure 1: On-demand prices by chip over time (left) and real prices against age (right). Source: the panel described in Section 3; outlier-flagged listings excluded.

4 Methodology

4.1 Hedonic rental-price model

For listing i of chip c in month t,

$$\ln p_{ict} = \alpha_c + g(a_{ct}) + \delta_t + X_i'\beta + \varepsilon_{ict}, \tag{1}$$

where p is the real posted price per GPU-hour, a_{ct} the chip's age in years since its release month, g the age profile, α_c a chip (vintage) effect, δ_t a month effect and X_i fixed effects for provider $\times$ region, contract type $\times$ term, GPUs per instance, price basis (instance or accelerator only) and form factor. g is a cubic spline in age with knots at 1.5, 3 and 5 years. The specifications were logged before any regression was run, together with four hypotheses. H1, front-loaded decline: value falls fastest in a chip's first two years, closer to geometric than to straight-line. H2, obsolescence shocks: a successor's release causes a discrete drop in the predecessor's rent. H3, contract-type wedge: spot and on-demand prices fall faster with age than reserved prices. H4, demand regimes: shortages mask age effects only temporarily, so age profiles are stable across sub-periods once month effects are included.

4.2 Identification

Because $a_{ct} = t - v_c$, any linear trend in age can be moved into the chip and month effects, so the linear part of g is not identified in (1) (Hall, 1968). Its curvature is identified, but only through differences in calendar-time trends across vintages: $a^2 = t^2 - 2vt + v^2$, and $-2vt$ is a chip-specific trend. I report five objects.

1. *Observed cohort paths.* Each chip's price path, by six-month age bin, with the controls but without month effects. This needs no identification assumption: it is what happened.

2. *Strategy 1, quality-adjusted vintage.* Replace α_c with observed specifications: (a) ln BF16 dense TFLOP/s alone, and (b) adding ln memory bandwidth and ln memory. The linear age term is then identified from the cross-chip relation between residual price and release date.

3. *Strategy 2, frontier-relative obsolescence.* Keep α_c, drop δ_t and add the chip's TFLOP/s relative to the best chip released by month t, after Geske et al. (2007). The assumption is that market-wide time effects work only through the frontier.

4. *Strategy 3, successor event study.* Event-time dummies around the release of each chip's successor (V100 at A100, A100 at H100, H100 at B200), with chip and month effects.

5. *Strategy 4, shape only.* Keep both sets of effects, identify only the curvature, and report $g_d(a) = d\,a + \tilde{c}(a)$ over a grid of assumed average rates d from 0 to -0.40 log points a year, where $\tilde{c}$ is the curvature detrended over ages 0–6.

The summary measures are $g(1) - g(0)$, $g(2) - g(0)$, $g(4) - g(0)$, the average annual log change over years 0–2 and 2–4, and their difference, the *early minus late rate*. H1 predicts a negative early minus late rate. Strategy 4 reports only that difference, the one measure that does not depend on the assumed linear term.

4.3 Inference

Age has one value per chip and month, so inference is at the chip level, with seven clusters. The method specified in advance is the unrestricted wild cluster bootstrap with Webb's six-point weights and 9,999 draws (Cameron et al., 2008; Webb, 2023). Cluster-robust standard errors by provider and two-way by provider and month are shown for comparison. After the adversarial review (Appendix B) I add the restricted bootstrap (WCR), which imposes the null when generating draws and is recommended when clusters are few and unequal in size (MacKinnon and Webb, 2017; MacKinnon et al., 2023), a test-inversion interval, and a Monte Carlo of interval coverage on this panel's design.

4.4 From rents to asset values

The value of a chip at age a is the present value of its remaining net rent,

$$V(a) = \sum_{m=0}^{12(H-a)-1} \frac{730\,\max\{u\,p(a + m/12) - o,\ 0\}\,S(m)}{(1+r)^{(m+1)/12}}, \tag{2}$$

summed over remaining months m at 730 hours a month, with utilization u, operating cost per GPU-hour o (system power $\times$ PUE $\times$ electricity price, plus maintenance at 4% of capital cost a year), survival S at a 2% annual failure rate, discount rate r and horizon H of nine years, the

oldest age observed. The rent path is $p(a) = p_{\text{now}} \exp\{g(a) - g(a_{\text{now}})\}$: each chip's current neocloud median on-demand rent moved along a profile g from the realized cohort paths or one of the strategies. I report $V(0)$ relative to capital cost, the share of value lost by each year, the economic life T^* (the first age at which $up \leq o$), and a straight-line-equivalent life $L = 2\int_0^H V(a)/V(0)\,da$, the straight-line life with the same average book value. The base case is $r = 10\%$, $u = 70\%$ and a PUE of 1.3; the grid is $r \in \{8, 10, 12\}\%$, $u \in \{50, 70, 90\}\%$ and PUE $\in \{1.2, 1.4\}$.

4.5 Earnings restatement

Section 6 enters each year's estimated GPU capital expenditure as a purchase cohort and depreciates it under the firm's disclosed straight-line life and under the rental-implied schedules from (2), with the GPU share of capital expenditure varied over a grid because no firm discloses it.

5 Results

5.1 Observed price paths

Figure 2 shows each chip's realized price path relative to its first observed six-month bin, holding provider, region, contract and the other controls fixed. These paths need no identification assumption. The older chips decline slowly and unevenly: the V100 changes by −0.13 log points over four years and −0.38 over nine, and the A100 40GB reaches −0.45 at five years. The H100's rent instead rose in its first year (0.24), fell to −0.50 at three years, and recovered to −0.04 at four years in the 2026 shortage. Realized paths are not smooth decays. They are long flat stretches, sharp falls and demand-driven rebounds, and the 2026 shortage reverses part of the decline for every chip from the A100 on, while the V100's path stays flat.

5.2 Age profiles under four identification strategies

Table 3 and Figure 3 report the strategies that identify a linear term, and the shape-only strategy. Strategy 1 prices each log point of TFLOP/s at 0.88 log points of rent. It finds no decline with age over the first two years (rate 0.27, interval [−0.23, 0.77]) and roughly none after (−0.03). Strategy 2's age profile rises slightly and then falls (0.11 and −0.15 a year). Strategy 4's shape estimate is 0.47 [0.36, 0.57] under the bootstrap specified in advance. Every point estimate of the early minus late rate is positive: prices fall later, not earlier, against H1.

Table 3: Summary of the age profile by identification strategy (full sample)

Strategy	$g(2) - g(0)$	$g(4) - g(0)$	rate, yrs 0–2	rate, yrs 2–4	early − late rate (H1: < 0)
1a quality index (TFLOP/s)	0.55 [−0.45, 1.55]	0.48 [−0.06, 1.03]	0.27 [−0.23, 0.77]	−0.03 [−0.24, 0.18]	0.31 [−0.52, 1.11]
1b three specs	0.55 [−0.57, 1.68]	0.55 [−0.61, 1.69]	0.28 [−0.28, 0.84]	0.00 [−0.50, 0.47]	0.28 [−0.42, 0.97]
2 frontier-relative	0.22 [−0.25, 0.69]	−0.08 [−0.77, 0.62]	0.11 [−0.12, 0.35]	−0.15 [−0.41, 0.11]	0.26 [0.05, 0.47]
4 shape only	–	–	–	–	0.47 [0.36, 0.57]

Log points; rates are per year. Brackets: 95% intervals from the unrestricted wild cluster bootstrap at the chip level (7 clusters, Webb weights, 9,999 draws). These intervals are too narrow; see Table 4.

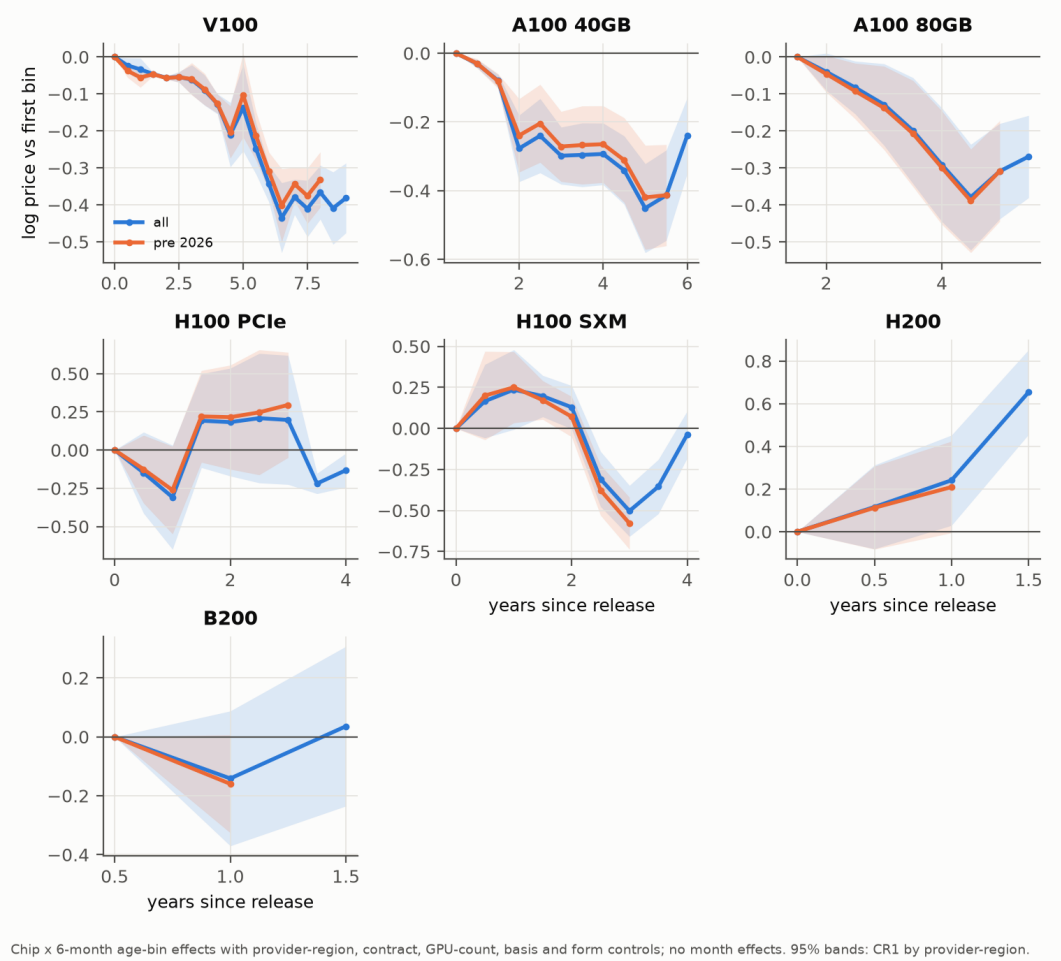


Figure 2: Observed cohort paths: log real price by six-month age bin relative to each chip's first bin, with provider × region, contract × term, GPU count, price basis and form-factor controls and no month effects. Bands: 95% intervals, CR1 by provider-region.

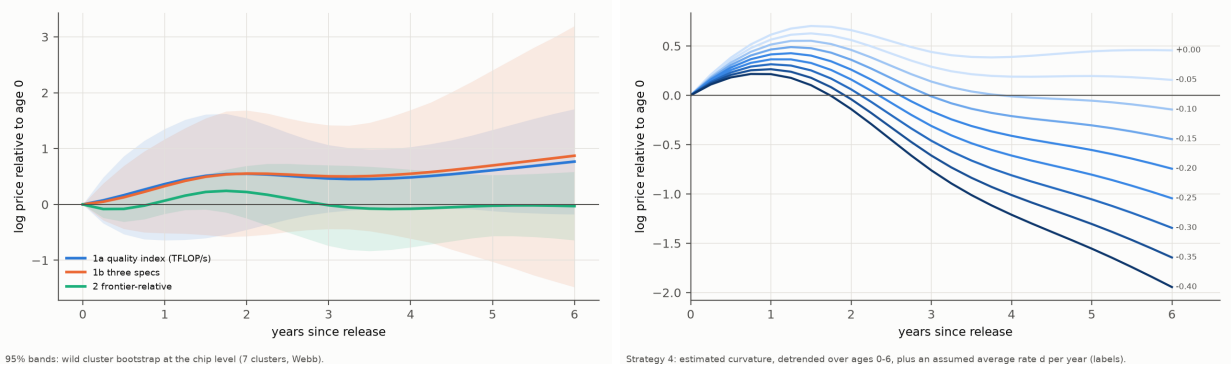


Figure 3: Age profiles $g(a) - g(0)$ under strategies 1a, 1b and 2 (left) and strategy 4 for assumed average rates d from 0 to -0.40 log points a year (right).

5.3 What survives valid inference

An adversarial review of these results (Appendix B), written by a reviewer with no access to the author's notes, identified three problems; Table 4 and the checks below address them.

The intervals are too narrow. In a Monte Carlo that keeps this panel's rows, fixed effects and regressors and draws errors with the estimated residual structure under a true shape of zero, the unrestricted bootstrap's 95% intervals cover the truth 66%–72% of the time and CR1 by chip 74%–76% (200 replications per design); the restricted bootstrap covers 98%–98%. Under the restricted bootstrap no summary measure is significant: strategy 4's shape has $p = 0.29$ (unrestricted p <0.0001) and a test-inversion interval of [−0.44, 1.03], and strategy 2's has $p = 0.30$.

The shape is identified only from vintage-specific trends. Allowing each chip its own linear calendar trend leaves the quadratic term unidentified; at a zero quadratic term the shape is −0.39. Reassigning release dates across chips, the actual assignment gives the largest shape of 1,260 distinct assignments ($p = 0.0008$), so the curvature is tied to release order. That test cannot tell an age effect from demand trends that are themselves ordered by vintage, such as the 2023–26 shortages that hit the newest chips.

The answer depends on the market. The shape is 0.87 for spot prices [0.34, 1.41] and −0.13 for on-demand prices [−0.40, 0.13]. Restricting to the 5,226 non-spot listings whose price changes at least once, it is −0.52 (provider-clustered standard error 0.24), in H1's direction. Weighting providers equally cuts the pooled shape to 0.19.

Table 4: Unrestricted (WCU) and restricted (WCR) wild cluster bootstrap p-values

Strategy	Measure	Estimate	SE (CR1, chip)	p (WCU)	p (WCR)
1a quality index	rate, yrs 0–2	0.27	0.16	0.300	0.649
1a quality index	rate, yrs 2–4	−0.03	0.06	0.615	0.678
1a quality index	early − late	0.31	0.19	0.421	0.735
1b three specs	rate, yrs 0–2	0.28	0.16	0.262	0.595
1b three specs	rate, yrs 2–4	0.00	0.14	0.991	0.993
1b three specs	early − late	0.28	0.22	0.591	0.820
2 frontier-relative	rate, yrs 0–2	0.11	0.09	0.297	0.494
2 frontier-relative	rate, yrs 2–4	−0.15	0.06	0.208	0.228
2 frontier-relative	early − late	0.26	0.10	0.006	0.305
4 shape only	early − late	0.47	0.08	<0.0001	0.287

Null: measure = 0. Chip-level clusters (7), Webb weights, 9,999 draws. SE: CR1 clustered by chip.

Taken together, the projection strategies do not establish either the speed or the shape of the age profile in this sample. The observed cohort paths are the more credible guide to the rate, and the dependence of the shape on the market read is itself a finding.

5.4 Successor releases

Figure 4 shows the predecessor's price around its successor's release. There is no detectable drop: prices in the first year after release are −0.09 [−0.40, 0.23] relative to the six months before. The

one negative bin, −0.18 at 6–11 months, falls where cloud availability of the successor actually began; dating events by the successor's first listing in the panel moves that bin earlier, and neither dating detects a drop (H2).

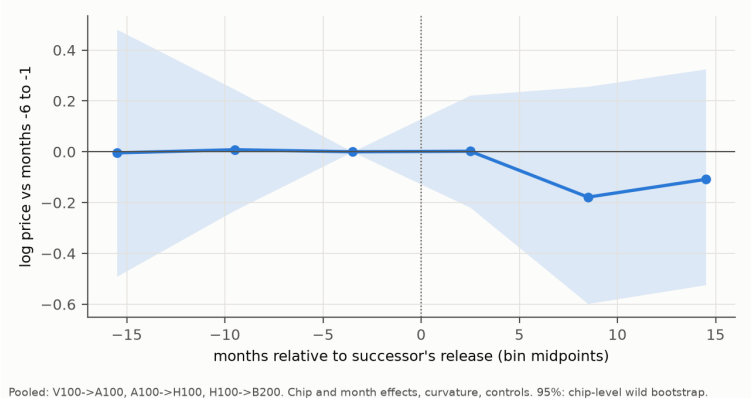


Figure 4: Successor event study: predecessor prices by six-month event-time bin around the successor's Epoch release month, relative to months −6 to −1; chip and month effects and controls. Bands: unrestricted chip-level bootstrap.

5.5 Asset values and economic life

Table 5 and Figure 5 turn the rent paths into values. Along the realized path an H100 loses 11% of its value in the first year and 32% by year three, and its straight-line-equivalent life is 9.4 years: value declines more slowly than under any five- or six-year straight-line schedule. Strategy 2 gives a similar curve (28% by year three). Under strategy 4 an assumed average decline of 0.20 log points a year gives 53% lost by year three and a life of 7.1 years, about the pace of a six-year straight-line schedule over the first three years. Only an assumed decline of 0.40 falls faster than a five-year schedule (72% lost by year three, life 4.9 years), and it implies a launch rent of $16.02 an hour for the H100, far above any rent in the panel.

Two features of these numbers matter more than their levels. First, T^* never binds within the nine-year horizon: the H100's neocloud rent of $4.50 an hour is many times its $0.30 of power and maintenance, so the end of the value curve is set by the horizon, not by rents reaching operating cost. With a six-year horizon the straight-line-equivalent life is 6.1 years, and with five years 5.1. Rents alone do not pin down economic life; the retirement decision does. Second, $V(0)$ is 2.9–6.3 times capital cost across the grid, which reflects scarcity rents and the omitted hosting costs, so the value-to-cost ratio should not be read as profitability.

5.6 Triangulation with benchmark, term-structure and retirement evidence

A companion analysis measures three other quantities for the same chips: a throughput benchmark priced at neocloud rents (P1), the decay implied by reserved-to-on-demand price ratios (P2), and a power-constrained retirement model in which an old chip is replaced when a newer one would

Table 5: Asset values and straight-line-equivalent lives

Chip	Rent path	$V(0)$/capex	Share lost by yr 3	SL-equivalent life (yrs)	T^* (yrs)
A100 80GB	Realized (V100, A100)	5.32 [3.41, 7.48]	0.32 [0.31, 0.33]	9.4 [9.2, 9.6]	>9
A100 80GB	2 frontier-relative	4.83 [3.07, 6.83]	0.28 [0.27, 0.30]	10.1 [9.8, 10.3]	>9
A100 80GB	4 shape, d = -0.20	9.08 [6.02, 12.48]	0.52 [0.52, 0.53]	7.2 [7.0, 7.3]	>9
A100 80GB	4 shape, d = -0.40	19.94 [13.58, 26.82]	0.71 [0.71, 0.72]	5.0 [4.9, 5.1]	>9
H100 SXM	Realized (V100, A100)	4.49 [2.89, 6.31]	0.32 [0.31, 0.33]	9.4 [9.2, 9.6]	>9
H100 SXM	2 frontier-relative	4.55 [2.91, 6.41]	0.28 [0.27, 0.30]	10.1 [9.9, 10.3]	>9
H100 SXM	4 shape, d = -0.20	5.89 [3.87, 8.11]	0.53 [0.52, 0.54]	7.1 [6.9, 7.3]	>9
H100 SXM	4 shape, d = -0.40	8.89 [6.00, 12.02]	0.72 [0.72, 0.74]	4.9 [4.8, 5.0]	>9
H200	Realized (V100, A100)	5.57 [3.60, 7.80]	0.32 [0.31, 0.33]	9.4 [9.2, 9.6]	>9
H200	2 frontier-relative	4.81 [3.08, 6.79]	0.28 [0.27, 0.30]	10.1 [9.9, 10.3]	>9
H200	4 shape, d = -0.20	4.00 [2.58, 5.57]	0.54 [0.53, 0.56]	7.0 [6.7, 7.2]	>9
H200	4 shape, d = -0.40	3.81 [2.49, 5.23]	0.76 [0.74, 0.79]	4.5 [4.1, 4.7]	>9
B200	Realized (V100, A100)	4.45 [2.87, 6.23]	0.32 [0.31, 0.33]	9.4 [9.2, 9.6]	>9
B200	2 frontier-relative	3.84 [2.45, 5.42]	0.28 [0.27, 0.30]	10.1 [9.8, 10.3]	>9
B200	4 shape, d = -0.20	3.18 [2.05, 4.44]	0.54 [0.53, 0.57]	7.0 [6.7, 7.1]	>9
B200	4 shape, d = -0.40	3.03 [1.97, 4.17]	0.76 [0.75, 0.80]	4.4 [4.0, 4.7]	>9

Base case ($r = 10\%$, $u = 70\%$, PUE 1.3); brackets: range over the $r \times u\times$ PUE grid. Realized: the cohort paths of the V100 and A100. d: assumed average rate in log points a year. $V(0)$/capex: value at release over capital cost. SL-equivalent life: $2\int_0^H V(a)/V(0)\,da$. T^*: first age at which utilization $\times$ rent $\leq$ operating cost. Operating cost excludes hosting, network and staff, so $V(0)$/capex is an upper bound.

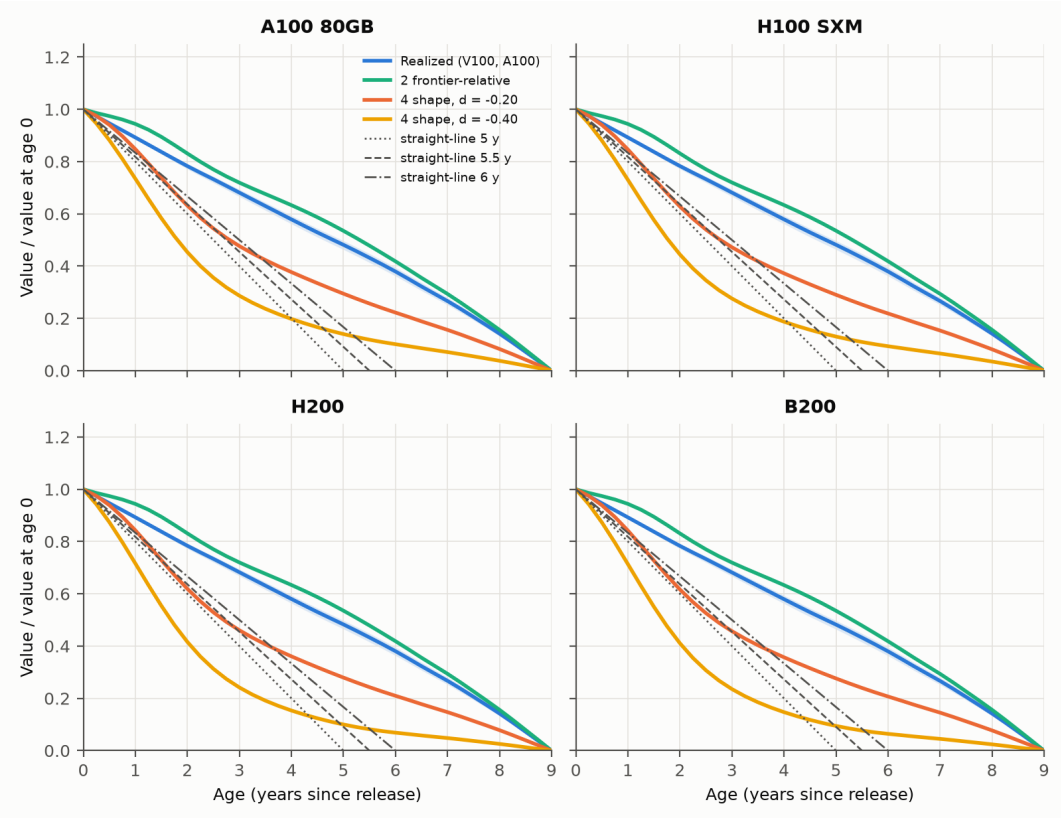


Figure 5: Value relative to value at release, by rent path, against 5-, 5.5- and 6-year straight-line schedules. Band: the $r \times u\times$ PUE grid around the realized path.

earn more in the same power slot (P3). Table 6 and Figure 6 compare them. Old chips are priced close to their output today (the H100's obsolescence discount against the H200 is −0.02). Life to operating cost is long: 9.8–11.7 years for the H100 from the term structure. In the power-constrained model the median retirement age is 4.1 years (P10–P90 2.2–7.0), rising to 11.0 years when power is abundant and 10.7 without a scarcity markup; about 47% of today's rent is markup at the calibrated level. Book lives of four to six years sit between the power-constrained and the unconstrained answers, close to the constrained one.

Table 6: Triangulation of economic life by method

Generation (age, yrs)	P1 discount D	P1 life to opex	P2 life to opex	P3 retirement age (P10–P90)
A100 80GB (6.0)	0.00 [−0.21, 0.20]	19.9–25.9	10.7–13.2	4.1 (2.2–7.0)
H100 SXM (4.1)	−0.02 [−0.13, 0.34]	15.1–20.4	9.8–11.7	4.1 (2.2–7.0)
H200 SXM (2.0)	0.00 [0.00, 0.00]	14.3–18.8	6.5–7.5	4.1 (2.2–7.0)

P1: obsolescence discount against the H200 at concurrency 32 (median neocloud price; provider range) and implied life to operating cost. P2: life to operating cost over the discount-rate and term-premium grid. P3: median retirement age, power-constrained base case. All lives in years since release.

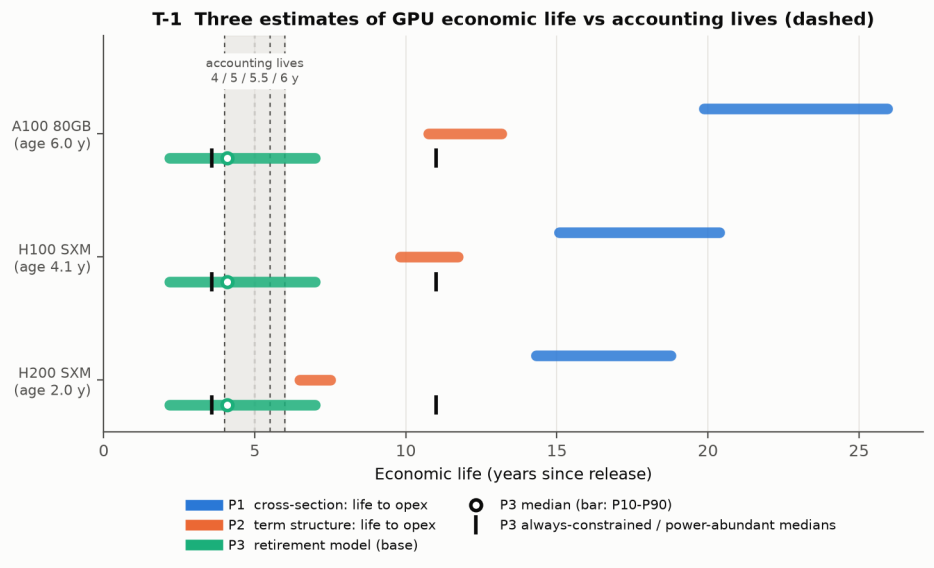


Figure 6: Economic life by method against accounting lives of 4, 5, 5.5 and 6 years.

6 Accounting implications

No firm reports GPUs separately, so the restatement estimates a GPU asset base. Each year's cash capital expenditure from XBRL is multiplied by the share that is servers and network equipment (0.6 in the baseline, 0.4 and 0.8 as alternatives; no firm discloses it) and by the GPU share of server spending (0.3 to 0.8). For Microsoft in fiscal 2026, for example, $115.9 billion of capital expenditure becomes a GPU base of $34.8 billion at a GPU share of one half. Each year's purchases form a cohort from fiscal 2020, depreciated under the firm's disclosed straight-line life (Table 1;

changes applied prospectively, as the firms did) and under each alternative schedule. The change in operating income is the negative of the change in depreciation; the change in diluted EPS applies the firm-year effective tax rate. Every firm-year number traces to an XBRL fact listed with its accession number in the replication package.

Table 7 reports the latest fiscal year for each firm. Under the rental-implied schedule built on the realized cohort paths, which depreciates slower than any of the disclosed lives, the five hyperscalers' operating income would be 2.6%–15.5% *higher* ($1.5–12.4 billion) at a GPU share of one half. Under the steepest schedule, strategy 4 with an assumed decline of 0.40 log points a year, the change ranges from −11.0% to 3.3%; the largest cut is Oracle's, −11.0% of fiscal 2026 operating income, or $−0.68 per diluted share. A four-year straight-line life gives a similar range (−10.4% to 1.9%). CoreWeave and Nebius, whose operating income is small relative to their GPU spending, are shown in dollars only.

Table 7: Change in operating income and diluted EPS under alternative GPU depreciation schedules, latest fiscal year

Firm	FY	SL 3y	SL 4y	SL 5y	SL 5.5y	SL 6y	M1	M2	M3	M4
Panel A: change in operating income, USD bn										
MSFT	2026	−4.80 to −12.79	−2.69 to −7.17	−1.27 to −3.39	−0.70 to −1.87	−0.23 to −0.60	+2.56 to +6.83	+3.45 to +9.20	+0.30 to +0.81	−2.55 to −6.80
GOOGL	2025	−3.59 to −9.56	−2.36 to −6.29	−1.38 to −3.67	−0.98 to −2.61	−0.65 to −1.73	+2.02 to +5.39	+2.76 to +7.36	+0.19 to +0.51	−2.07 to −5.52
AMZN	2025	−0.24 to −0.65	+0.91 to +2.42	+1.69 to +4.50	+2.32 to +6.18	+2.84 to +7.57	+7.44 to +19.84	+8.51 to +22.70	+4.74 to +12.63	+1.59 to +4.23
META	2025	−2.46 to −6.56	−1.86 to −4.96	−1.04 to −2.77	−0.68 to −1.82	−0.39 to −1.03	+1.71 to +4.56	+2.29 to +6.12	+0.34 to +0.90	−1.35 to −3.61
ORCL	2026	−2.15 to −5.74	−1.29 to −3.43	−0.62 to −1.65	−0.34 to −0.89	−0.10 to −0.27	+0.87 to +2.32	+1.35 to +3.59	−0.01 to −0.03	−1.36 to −3.63
CRWV	2025	−0.66 to −1.76	−0.33 to −0.88	−0.13 to −0.35	−0.06 to −0.16	−0.00 to −0.00	+0.22 to +0.58	+0.30 to +0.79	−0.06 to −0.16	−0.38 to −1.02
NBIS	2025	−0.07 to −0.20	−0.00 to −0.00	+0.04 to +0.12	+0.06 to +0.16	+0.07 to +0.20	+0.12 to +0.33	+0.16 to +0.42	+0.07 to +0.20	−0.02 to −0.05
Panel B: change in diluted EPS, USD (effective tax rate)										
MSFT	2026	−0.52 to −1.38	−0.29 to −0.78	−0.14 to −0.37	−0.08 to −0.20	−0.02 to −0.07	+0.28 to +0.74	+0.37 to +1.00	+0.03 to +0.09	−0.28 to −0.73
GOOGL	2025	−0.24 to −0.65	−0.16 to −0.43	−0.09 to −0.25	−0.07 to −0.18	−0.04 to −0.12	+0.14 to +0.37	+0.19 to +0.50	+0.01 to +0.03	−0.14 to −0.38
AMZN	2025	−0.02 to −0.05	+0.07 to +0.18	+0.13 to +0.33	+0.17 to +0.46	+0.21 to +0.56	+0.55 to +1.47	+0.63 to +1.69	+0.35 to +0.94	+0.12 to +0.31
META	2025	−0.67 to −1.79	−0.51 to −1.36	−0.28 to −0.76	−0.19 to −0.50	−0.11 to −0.28	+0.47 to +1.25	+0.63 to +1.67	+0.09 to +0.25	−0.37 to −0.99
ORCL	2026	−0.65 to −1.72	−0.39 to −1.03	−0.19 to −0.49	−0.10 to −0.27	−0.03 to −0.08	+0.26 to +0.70	+0.40 to +1.08	−0.00 to −0.01	−0.41 to −1.09
CRWV	2025	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
NBIS	2025	−0.21 to −0.56	−0.00 to −0.00	+0.13 to +0.34	+0.17 to +0.47	+0.21 to +0.57	+0.35 to +0.94	+0.45 to +1.21	+0.21 to +0.56	−0.06 to −0.15

Each cell: GPU share of server capex 0.3 to 0.8; server/network share of capex 0.6 (assumption); cohorts from FY2020. Change = alternative schedule minus the firm's disclosed straight-line book life. n/a: effective tax rate undefined (pre-tax loss). M1: `market_realized_older_h100_sxm` (Phase 6). M2: `market_s2_frontier_h100_sxm` (Phase 6). M3: `market_s4_d-0.20_h100_sxm` (Phase 6). M4: `market_s4_d-0.40_h100_sxm` (Phase 6).

M1 to M4 are the rental-implied schedules for the H100 from Table 5: M1 the realized cohort paths, M2 strategy 2, M3 and M4 strategy 4 at assumed average declines of 0.20 and 0.40 log points a year.

The sign of the restatement depends on the rent path, not on the method. Posted rents, read through the realized paths, imply that book lives are, if anything, conservative. Only an assumption of steep age-driven decline, which the rent data do not establish, produces the large earnings overstatements discussed in the public debate. The restatement inherits the limits of the asset base: cash capital expenditure excludes finance-lease additions, which are material for Amazon and Microsoft, and the server and GPU shares are assumptions shown as a grid.

7 Robustness and limitations

Contract type and sub-period. Table 8 repeats the strategies by contract type (H3) and sub-period (H4). The contract split is the important one (Section 5.3). The sub-period split cannot test H4 as intended: 2026 alone covers under a year of each chip's life, which is why its intervals are uninformative, and before 2023 only three chips are on sale, too few clusters to report.

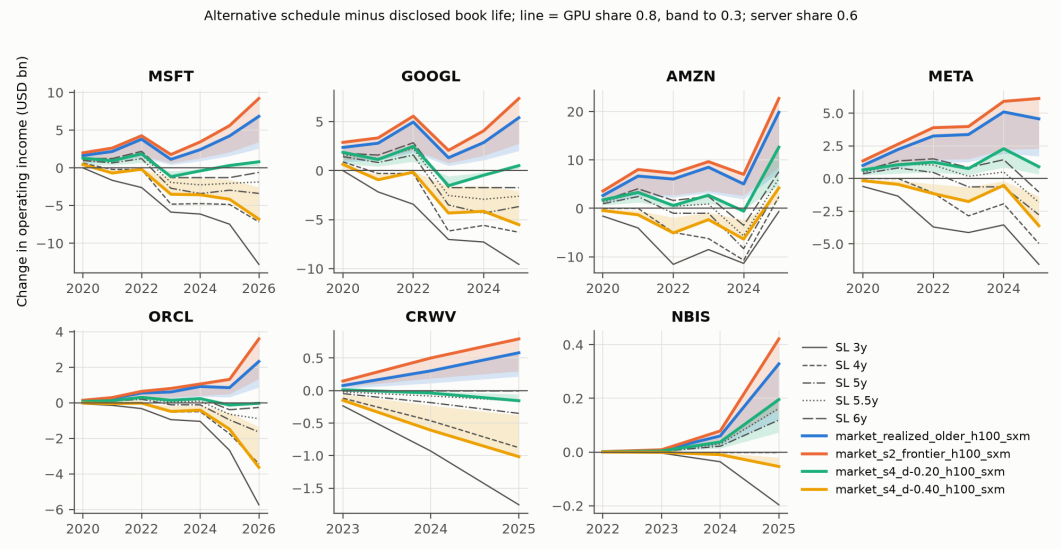


Figure 7: Change in operating income by firm and schedule across the GPU-share grid.

Table 8: Age profile by contract type and sub-period

Sample	Strategy	rate, yrs 0–2	rate, yrs 2–4	early − late (H1: < 0)
On demand	1a quality index (TFLOP/s)	−0.07 [−0.47, 0.34]	0.10 [−0.16, 0.37]	−0.17 [−0.70, 0.37]
On demand	2 frontier-relative	−0.21 [−0.48, 0.07]	−0.09 [−0.36, 0.17]	−0.12 [−0.33, 0.08]
On demand	4 shape only	–	–	−0.13 [−0.40, 0.13]
Reserved, 1 year	1a quality index (TFLOP/s)	0.81 [−0.54, 2.19]	0.18 [−0.08, 0.45]	0.63 [−0.78, 2.03]
Reserved, 1 year	2 frontier-relative	0.08 [−0.18, 0.34]	−0.06 [−0.23, 0.11]	0.14 [−0.23, 0.50]
Reserved, 1 year	4 shape only	–	–	0.76 [−1.15, 2.66]
Reserved, 3 years	1a quality index (TFLOP/s)	0.19 [−0.07, 0.45]	0.10 [−0.18, 0.37]	0.09 [−0.05, 0.24]
Reserved, 3 years	2 frontier-relative	−0.03 [−0.07, 0.01]	−0.12 [−0.28, 0.04]	0.09 [−0.13, 0.31]
Reserved, 3 years	4 shape only	–	–	−0.20 [−0.80, 0.38]
Spot	1a quality index (TFLOP/s)	0.46 [0.11, 0.81]	−0.19 [−0.32, −0.04]	0.65 [0.19, 1.10]
Spot	2 frontier-relative	0.50 [0.18, 0.83]	−0.26 [−0.42, −0.09]	0.76 [0.26, 1.26]
Spot	4 shape only	–	–	0.87 [0.34, 1.41]
Without Google, July 2026	1a quality index (TFLOP/s)	0.28 [−0.21, 0.76]	−0.03 [−0.24, 0.18]	0.31 [−0.50, 1.09]
Without Google, July 2026	2 frontier-relative	0.11 [−0.12, 0.35]	−0.15 [−0.39, 0.10]	0.26 [0.05, 0.47]
Without Google, July 2026	4 shape only	–	–	0.47 [0.36, 0.57]
2023–2025	1a quality index (TFLOP/s)	0.26 [−0.11, 0.63]	−0.27 [−0.43, −0.12]	0.54 [0.19, 0.88]
2023–2025	2 frontier-relative	0.11 [−0.09, 0.32]†	−0.37 [−1.62, 0.86]†	0.48 [0.34, 0.63]
2023–2025	4 shape only	–	–	0.48 [0.28, 0.68]
2026	1a quality index (TFLOP/s)	−0.36 [−53.25, 53.09]	−0.09 [−0.29, 0.12]	−0.27 [−52.97, 52.98]
2026	2 frontier-relative	−6.60 [−47.69, 35.77]†	0.82 [−2.86, 4.12]†	−7.42 [−44.76, 30.91]
2026	4 shape only	–	–	−0.61 [−11.87, 10.71]
Provider-chip-months weighted equally	1a quality index (TFLOP/s)	0.21 [−0.21, 0.62]	−0.03 [−0.18, 0.11]	0.24 [−0.15, 0.64]
Provider-chip-months weighted equally	2 frontier-relative	0.10 [−0.22, 0.43]	−0.08 [−0.17, 0.01]	0.19 [−0.11, 0.47]
Provider-chip-months weighted equally	4 shape only	–	–	0.29 [0.07, 0.50]

As Table 3. "Without Google, July 2026" drops one anomalous Google Cloud catalog snapshot. †: within-chip age span under four years with chip effects (extrapolated). Sub-samples with fewer than six chips are not reported.

Weighting and composition. Google Cloud is 62% of the estimation sample. Weighting each provider equally lowers the pooled shape to 0.19, and dropping Google leaves 0.44. Strategy 1's early-years rate depends on the H200: dropping it moves the rate from 0.27 to −0.11.

Posted versus transacted prices. The panel is posted prices, overwhelmingly hyperscaler list prices, and 96% of on-demand and reserved month-to-month transitions are unchanged. Large buyers pay negotiated prices that are not observed. The results are therefore about the posted-price market; a transacted-price panel, such as the marketplace data of Huang (2026), would be the natural complement.

Age zero. Age is measured from Epoch's release month, which precedes the first cloud listing by several months for some chips. The event study's sensitivity to that dating is reported in Section 5.4.

Operating cost. Hosting, network and staff costs are excluded for lack of a sourced figure, so values are upper bounds and T^* is biased upward.

8 Conclusion

Published estimates of accelerator depreciation disagree because they read different prices through different assumptions. In a panel of posted rents for seven NVIDIA chips, realized rents fall slowly; the shape of the age profile flips between spot and list prices; and with valid few-cluster inference no identification strategy pins down the rate or the shape. Rents stay far above operating cost for the whole observed life, so the economic life that matters for book depreciation is decided by obsolescence in power-constrained data centers, not by rents reaching operating cost.

For accounting, the implication is narrower than either side of the public debate. Five- or six-year straight-line lives are not contradicted by posted rents, whose value curves decline more slowly than straight-line. They look long only if one believes that power-constrained operators retire chips when a better chip arrives, which is what the retirement model and the largest operators' own behavior suggest; even then the model's median retirement age of 4.1 years has a P10–P90 range of 2.2–7.0 years that contains those lives. The firm-level effect of that choice is reported in Section 6.

The panel, code and identification range are public, so each of these choices can be revisited as longer price histories for the Hopper and Blackwell generations accumulate.

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A Data and code availability

The replication package rebuilds every table and figure, and this paper, from the archived raw files with one command (`make all`). Each raw file has a sidecar recording its source URL, retrieval time and HTTP status. Prices from providers whose terms do not permit redistribution are released as derived monthly aggregates; the SkyPilot catalog history and archived pages are re-fetched by a script with checksums rather than redistributed. Methodological choices are logged with dates and alternatives in `docs/decisions.md`, and every assumption with its value and source in `docs/assumptions.md`.

B Adversarial review

Before the results were written up, a reviewer with access only to the code, the result tables and the identification section of the research proposal was asked for the weakest identifying assumption and the most likely referee objection. Its fifteen points, the checks they led to and the responses are in `docs/review_01.md` and `docs/review_01_checks.md`; the checks are in `gel.models.review_checks` and Section 5.3.

C Statements

Use of AI tools. This study was carried out with Claude, a language model made by Anthropic, running in Claude Code. Under the author's direction, Claude wrote the collectors, parsers and estimation code, ran the analyses, verified the references against their sources, produced the figures and tables, and drafted the text of this paper; a separate instance with no access to the author's notes wrote the adversarial review in Appendix B. The author set the research questions and hypotheses, made or approved the methodological decisions logged in the replication package, and takes responsibility for the content. No number in the paper is typed by hand: each is generated by the replication package's code from archived data.

Declaration of interests. None to declare.

Funding. This research received no external funding.

Licences. Code is released under the MIT licence; the text and the released dataset under CC BY 4.0. Third-party data keep their sources' terms (Appendix A).